GABRIEL AFRIAT

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EDUCATION

Massachusetts Institute of Technology Cambridge, MA
PhD Candidate in Operations Research, Advisor: Prof. Rahul Mazumder, GPA: 5.0/5.0 2022 – Present

- *Research Focus:* Compression of LLMs, VLMs, computer vision, and recommendation models (pruning, quantization), robust optimization, anomaly detection and correction in large scale datasets, additive models
- *Relevant Coursework:* Nonlinear Optimization, Integer Optimization, Statistical Learning Theory and Applications, Mathematical Statistics: A Non-Asymptotic Approach, Sensorimotor Learning

Master of Business Analytics, MIT Sloan and Operations Research Center, GPA: 5.0/5.0 2021 – 2022

- *Relevant Coursework:* ML under a Modern Optimization Lens, Robust Optimization, Advances in Computer Vision
- *Awards:* Jean Gaillard Memorial Fellowship from Harvard University

CentraleSupélec (Paris-Saclay University) Gif-sur-Yvette, France
Master of Science in Engineering, Computer Science, GPA: 4.25/4.33 2019 – 2021
Bachelor of Science in Engineering, GPA: 4.33/4.33

- *Relevant Coursework:* Advanced Statistics, Artificial Intelligence, Statistics and Learning, Algorithmics and Complexity
- *Bachelor Class Rank:* Top 1.5%

Lycée Henri IV Paris, France
Preparatory classes in mathematics and physics for the Grandes Écoles national competitive exam 2016 – 2019

- *Relevant Coursework:* Linear Algebra, Abstract Algebra, Topology, Probability, Mathematical Analysis, Multivariable Calculus

RESEARCH AND INDUSTRY EXPERIENCE

Massachusetts Institute of Technology Cambridge, MA
Doctoral Research Assistant to Prof. Rahul Mazumder Jun 2022 – Present

- Exploring new methods for LLM/VLM compression (input compression, weight pruning, non-uniform sparsity allocation)
- Investigating new algorithms for multi-modal error detection and error correction
- Developed a new efficient framework for GAMs with interactions using sparsity and structural interpretability constraints

Google – YouTube San Bruno, CA
Student Researcher Jun 2026 – Aug 2026

- Worked on advanced quantization techniques for recommender models

Google Research New York, NY
Student Researcher Jun 2024 – Aug 2024

- Developed RobOP (ICML ’26), a robust optimization framework for pruning vision transformers and LLMs without retraining
- Derived tractable robust pruning formulations that extend existing SOTA pruning methods (ALPS, CAP) to account for calibration noise and Hessian uncertainty (concentration bounds on Hessian estimate, stale gradients, etc.)
- Improved ImageNet-1K accuracy by up to 10 points and reduced WikiText-2 perplexity by up to 10 points over baseline

Massachusetts Institute of Technology / Grab Cambridge, MA
Research Capstone Project, advised by Prof. Daniel Freund and Prof. Stephen Graves Feb 2022 – Aug 2022

- Formulated a robust and distributionally robust optimization algorithm for pickup/delivery VRP with travel time uncertainty
- Developed a new approach for clustering, graph pruning and used warm start construction to improve algorithm efficiency
- Reduced maximum order delay by 70% and the proportion of delayed orders by up to 16% on the test set from Grab

Massachusetts Institute of Technology Cambridge, MA
Research Assistant to Prof. Dimitris Bertsimas Sep 2021 – May 2022

- Proposed a new way to model node sparsity constraints in neural networks to reproduce the brain's capacity for specialization
- Explored the use of Optimal Decision Trees to improve interpretability of Reinforcement Learning agents

PUBLICATIONS

Conference and Journal Papers

TMLR 2026 *MOONSHOT: A Framework for Multi-Objective Pruning of Vision and Large Language Models*
Gabriel Afriat, Xiang Meng, Shibal Ibrahim, Hussein Hazimeh, Rahul Mazumder

ICML 2026 *A Robust Optimization Guided Pruning Framework for Vision and Large Language Models*
Gabriel Afriat, Hussein Hazimeh, Dimitris Paparas, Rahul Mazumder

ICLR 2026 *Seeing What's Wrong: A Trajectory-Guided Approach to Caption Error Detection*
Gabriel Afriat, Ryan Lucas, Xiang Meng, Yufang Hou, Yada Zhu, Rahul Mazumder

KDD 2025 *An Optimization Framework for Differentially Private Sparse Fine-Tuning*
Mehdi Makni, Kayhan Behdin, **Gabriel Afriat**, Zheng Xu, Sergei Vassilvitskii, Natalia Ponomareva, Rahul Mazumder, Hussein Hazimeh

NeurIPS 2023 *GRAND-SLAMIN' Interpretable Additive Modeling with Structural Constraints*
Shibal Ibrahim, **Gabriel Afriat**, Kayhan Behdin, Rahul Mazumder

Workshop Papers

NeurIPS 2026 *ViPADIA: A Lightweight Robust Visual Prompting Machine with applications to Industrial Anomaly Detection and Reasoning*
Gabriel Afriat, Naresh Iyer, Bechir El Hannachi, Rahul Mazumder. *Workshop VLM4RWD*

TEACHING

Massachusetts Institute of Technology Cambridge, MA
Teaching Assistant, MIT Sloan School of Management 2023 – 2025

- The Analytics Edge (15.071), ML class with 100+ students — Fall 2023, Spring 2025
- Advanced Analytics Edge (15.072), ML class with 80+ students — Fall 2023, Fall 2024
- Hands-on Deep Learning (15.773), ML class with 80+ students — Spring 2024

CentraleSupélec (Paris-Saclay University) Gif-sur-Yvette, France
Teaching Assistant 2020 – 2021

- Convergence, Integrability and Probability (1SL1000) — Fall 2020
- Partial Differential Equations (1SL1500) — Spring 2021

SKILLS AND ADDITIONAL INFORMATION

Programming Python, R, Julia, SQL, Git, OCaml, LaTeX, PowerShell
Software Tools PyTorch, JAX, TensorFlow, Remote Clusters, Distributed and Parallel Training, JuMP, Gurobi
Languages French (native), English (fluent), German (basic)
Hobbies Badminton, cooking, piano and guitar